

Technical Notes

Numerical Experiments on Anomalies from Stationary, Slowly Moving, and Fast-Moving Shocks

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DOI: 10.2514/1.J057366

Nomenclature

c_p	=	specific heat at constant pressure
E	=	total energy per unit mass
$\mathbf{E}, \mathbf{F}$	=	(inviscid) flux vectors in x and y directions
H	=	total enthalpy
M	=	(shock intensity) Mach number, $M \in (1, \infty)$
M_s	=	(shock propagation) Mach number, $M_s \in [0, \infty)$
M_{signal}	=	("signal") Mach number, $M_{\text{signal}} \approx 0.44M$ for perfect-gas-modeled air
p	=	pressure
T	=	temperature
u, v	=	velocity components in Cartesian coordinates
x, y	=	Cartesian coordinates
γ	=	specific heat ratio, 1.4
ϵ	=	shock location parameter, $\epsilon \in [0, 1]$
ϵ_g	=	grid perturbation parameter, 0, $1.e - 6$, or $1.e - 3$
ρ	=	density

Subscripts

L, R	=	left and right states of shock
0	=	stagnation value
∞	=	freestream value

I. Introduction

HYPersonic flow computations still suffer from shock-wave anomalous solutions, such as a "carbuncle phenomenon" (Fig. 1) [1–4]. They can severely degrade the predictability of hypersonic heating, which can be disastrous to space vehicles at reentry [5]. Let us briefly look back how we have derived numerical expressions for shock waves (Fig. 2): 1) in the real world, the shock is very thin, but has a certain thickness (physically) (Fig. 2a); 2) if we

express it by Euler equations (mathematically), on the other hand, we regard the shock as a discontinuous jump with zero thickness (Fig. 2b); and 3) however, when the shock is captured by a finite volume method (FVM) (numerically), it should have a certain thickness of the order of the cell size (Fig. 2c). It should be cautioned that this shock thickness, along with its internal structure, is a numerical artifact, not a physical one. Within the numerical shock, the continuity equation deviates from the Hugoniot curve up to 40% at Mach 10, for instance [6]. Therefore, we take the viewpoint that the shock anomalies are caused in part by the lack of mathematical expression for internal shock structure [6–8], and hence, we think that they should be examined by numerical experiments rather than (invalid) mathematical considerations.

Quirk [1] introduced a benchmark test for numerical schemes on their responses to the captured shock waves. His test employed a particular speed of a (fast) moving shock on a two-dimensional (2-D) perturbed grid. In his test, the shock took all the possible locations within a cell, but instantly passed through them. Roberts [9], on the other hand, chose a more slowly moving shock, which took 50 time steps to travel through a single cell when the Courant–Friedrichs–Lewy (CFL) number is unity, and discussed postshock numerical noise propagation. (The terms *slowly* and *fast-moving shocks*, to be redefined later here, were used differently in literature: the slowly moving shock meant that the shock travels a cell with huge time in [9,10], whereas it stood for across which shock characteristics change their signs in [11,12].) References [10–13] also investigated moving shocks and postshock oscillations both mathematically and numerically, typically in one dimension.

Kitamura et al. [3], inspired by [1,5,14,15], dealt with a stationary shock located within a row of cells in a 2-D (unperturbed) grid, with an initial shock position parameter $\epsilon = 0.0, 0.1, \dots, 0.9$, that is, the shock was placed at one of 10 possible locations in a cell. As stated in [12], there must be at least one cell to express a shock unless it is stationary and precisely located at the cell interface ($\epsilon = 0.0$). Thus, this set of shock locations represents every instant in which a moving shock experiences. The study in [3] discovered that one flux function has its own favorite shock location ϵ , and that such ϵ differs from one flux to another (Fig. 3). They also revealed in Refs. [3,16] that any flux functions, including flux-vector splitting (FVS) by van Leer [17], can generate carbuncles, although some of those methods had been believed to be carbuncle free according to Quirk's test [1] and others. In other words, Quirk's test for a moving shock does not support the robustness of a numerical flux for a stationary shock, in a strict sense. Therefore, a broader map that contains both [1] and [3] is desired as a systematic and thorough benchmark.

The present study will propose such a test by filling the gaps between the works explained earlier [1,3], that is, by varying the shock-propagation Mach number from 0 (of Kitamura et al. [3]) to 6 (in Quirk's choice [1]) and beyond, as the first attempt. Then, the effects of both shock intensity and shock-propagation Mach number will be discussed. Following our earlier work, eight different flux functions, such as by Roe [18], van Leer [17], Liou {advection upstream splitting method (AUSM)⁺-up [19]}, and Shima and Kitamura {simple low-dissipation AUSM (SLAU) [20]} will be used, because they have different degrees of robustness against the shock [16].

This Note is composed of the following: Sec. II will briefly explain the governing equations and numerical methods; Sec. III will present the numerical setup, which also states similarities and differences between [1] and [3]; Sec. IV will show numerical results; and Sec. V will conclude the current work.

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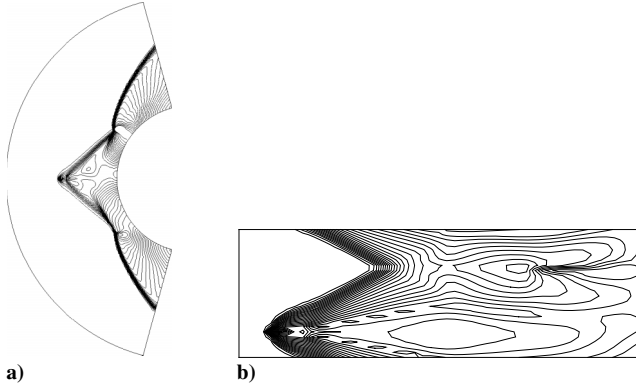


Fig. 1 Carbuncle solutions: a) blunt body and b) computational domain composed of squares.

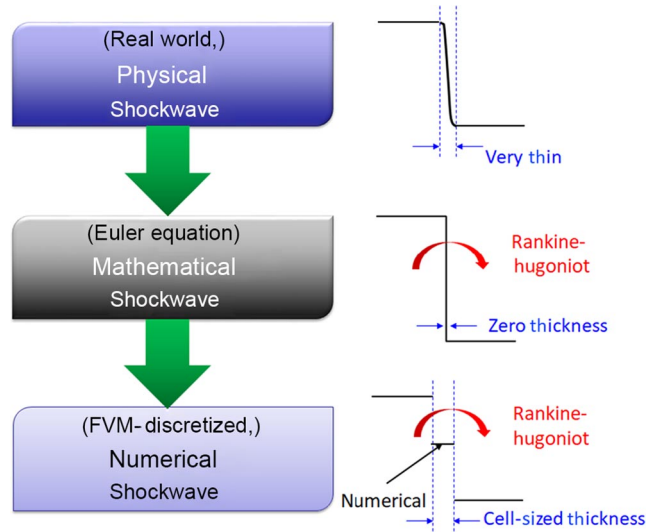


Fig. 2 Schematics of physical, mathematical, and numerically expressed shock waves.

II. Numerical Methods

A. Governing Equations

The 2-D Euler equations are solved in this study.

$$\frac{\partial Q}{\partial t} + \frac{\partial E}{\partial x} + \frac{\partial F}{\partial y} = 0 \quad (1a)$$

$$Q = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \quad E = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uH \end{bmatrix}, \quad F = \begin{bmatrix} \rho v \\ \rho vu \\ \rho v^2 + p \\ \rho vH \end{bmatrix} \quad (1b)$$

in which ρ is density; u, v are velocity components in Cartesian coordinates; E is total energy per unit mass; p is pressure; H is total

enthalpy [$H = E + (p/\rho)$]; and T is temperature. The working gas is assumed to be air approximated by the calorically perfect gas model with the specific heat ratio $\gamma = 1.4$.

B. Computational Method

The equations are discretized by a cell-centered FVM with first order both in space and time. We avoided using second- or higher-order schemes, because they reportedly tend to exhibit postshock oscillations from a moving shock more likely [9], whereas less carbuncles are produced from a stationary shock [3]—all these will possibly make our discussions more complicated. In addition, a numerical robustness of a second-order scheme against shock anomalies varied depending on a selected slope limiter in [21].

The numerical flux functions employed here are listed as follows.

1. Roe with Entropy Fix

In terms of robustness against shock, this flux function [18,22] is vulnerable to shock anomalies, and represents Godunov [23] or Harten–Lax–van Leer with contact [24] Riemann solver according to [3,16].

2. Van Leer's FVS

This flux function is very robust against shock, as well as Hänel et al. [25], Harten–Lax–van Leer–Einfeldt (HLLC) [26], and SLAU2 [27], although they (except for SLAU2) are too diffusive at contact discontinuity or boundary layer. It is modified later by Liou [28] to AUSM family.

3. AUSM⁺-up

Liou [19] developed the AUSM family numerical fluxes. The AUSM⁺-up [19], improved further from AUSM⁺, is employed here with tunable parameters, $K_\mu = 0.75$, $K_p = 0.25$, and $\sigma = 1.0$, as usual. In this paper, f_a is taken as unity, simply because only hypersonic flows are treated (see [19] for details).

4. SLAU

SLAU [20], developed by Shima and Kitamura, is one of the AUSM family schemes for all speeds. This flux is as moderately robust against the shock as AUSM⁺-up.

5. Low-Diffusion Flux-Splitting Scheme

Low-diffusion flux-splitting scheme (LDFSS) [29], proposed by Edwards, is also one of the AUSM family schemes. It is also as robust as AUSM⁺-up.

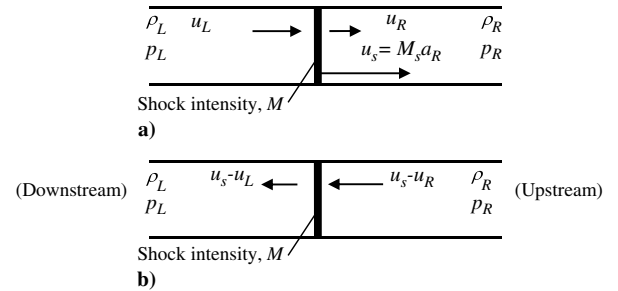


Fig. 4 Schematic of pre- and postshock relations: a) laboratory and b) shock coordinate systems.

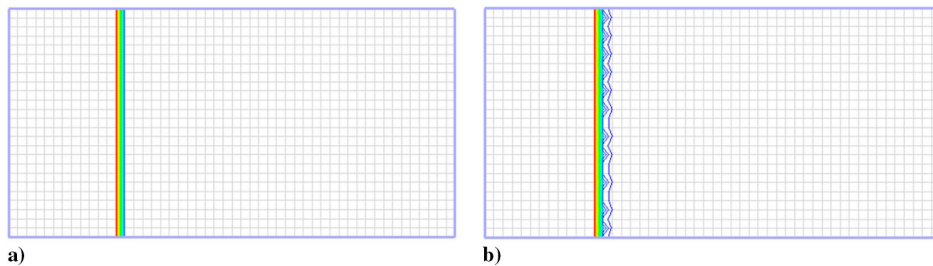


Fig. 3 Results of a) $\epsilon = 0.0$ (stable) and b) $\epsilon = 0.5$ (oscillatory) ($M_s = 0$; $M = 6$).

6. AUSM by Pressure-Based Weight Functions+

AUSM by pressure-based weight functions+ (AUSMPW+) [30] was developed by Kim et al. [31], from AUSM⁺. With the help of its multidimensional dissipation term, it is very robust as van Leer's FVS [17].

7. RoeM2

Kim et al. combined Roe and HLLE fluxes into RoeM2 [31]. This is again as robust as van Leer's FVS [17], thanks to its multidimensional term.

8. SLAU2

Kitamura and Shima enhanced SLAU in its robustness against shock anomalies in SLAU2 [27]. This is also very robust, although without a multidimensional dissipation.

After the numerical fluxes through all the cell interfaces are obtained, those contributions are integrated from cell to cell in the time evolution process, using the first-order Euler explicit method. Detailed information of the solver with regard to formulations and discretizations is found in [3,32].

in which u_s is shock-propagation velocity and expressed by the upstream condition as

$$u_s = M_s a_R, \quad a_R = \sqrt{\gamma p_R / \rho_R} \quad (4)$$

in which M_s is the (moving) shock Mach number [21], apparently, $u_s = 0$ for a stationary shock. [If $u_R = 0$ (the right region gas is at rest), M_s is identical to M and those relations reduce to simpler forms as in [25].]

B. Postshock Signal Mach Number

Here, let us consider the relation between the wave propagation speed a_L downstream of the shock (Fig. 5) and the a_R of upstream sound speed. After some manipulation, a_L is written as follows:

$$a_L = \sqrt{\gamma p_L / \rho_L} = a_R \cdot \sqrt{\left(\frac{p_L}{p_R}\right) / \left(\frac{\rho_L}{\rho_R}\right)} = a_R \cdot M_{\text{signal}} \quad (5a)$$

$$M_{\text{signal}}(M) \equiv \sqrt{\frac{1 + (2\gamma/(\gamma+1))(M^2-1)}{[1 + ((\gamma+1)/(\gamma-1))(1 + (2\gamma/(\gamma+1))(M^2-1))]/[(\gamma+1)/(\gamma-1) + (1 + (2\gamma/(\gamma+1))(M^2-1))]} \quad (5b)$$

$$\xrightarrow{M \rightarrow \infty} \frac{\sqrt{2\gamma(\gamma-1)}M}{\gamma+1} \approx 0.44M \quad (\text{if } \gamma = 1.4)$$

III. Numerical Experiment Setup

A. Basic Pre- and Postshock Relations

Let us consider a one-dimensional (1-D) inviscid flow setup. The flow across a normal shock wave experiences adiabatic changes, in which the basic equations, called Rankine–Hugoniot relations, are valid as follows [33] (Figs. 4a and 4b):

$$\frac{p_L}{p_R} = 1 + \frac{2\gamma}{\gamma+1}(M^2-1) \quad (2a)$$

$$\frac{\rho_L}{\rho_R} = \left[1 + \frac{\gamma+1}{\gamma-1}\left(\frac{p_L}{p_R}\right)\right] / \left[\frac{\gamma+1}{\gamma-1} + \left(\frac{p_L}{p_R}\right)\right] \quad (2b)$$

in which R and L are pre- (upstream) and postshock (downstream) states, respectively, in the shock coordinate system (Fig. 4b). M corresponds to upstream Mach number for a stationary shock, or relative upstream Mach number for a moving shock in the shock coordinate, which is regarded as shock “intensity” Mach number, $M = (u_s - u_R)/a_R$ (Fig. 4b). Then, the velocity normal to the shock is related by the continuity equation as

$$u_L = u_s \left[1 - \frac{\rho_R}{\rho_L}\right] + \frac{\rho_R u_R}{\rho_L} \left(\because \frac{u_s - u_L}{u_s - u_R} = \left(\frac{\rho_R}{\rho_L}\right)\right) \quad (3)$$

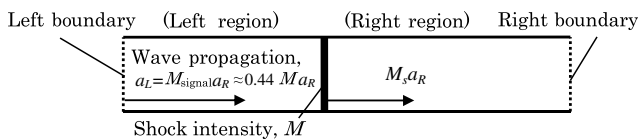


Fig. 5 Schematic of wave propagation, and left and right regions (laboratory coordinate system).

in which M_{signal} is defined as a signal Mach number, which is a function of only one parameter (i.e., the shock intensity M). It is interesting to observe that M_{signal} asymptotes to $0.44M$ for $M \gg 1$ (Fig. 6). [Note that $(a_L + |u_L|) > M_s a_R$ is *always* true, and $a_L \approx M_{\text{signal}} a_R$. However, it depends on the setup which of M_s or M_{signal} is greater.] Then, we can compare M_s with M_{signal} (or $0.44M$, instead): if $M_{\text{signal}} > M_s$, numerical noise in the left (downstream) region catches up with the moving shock and can disturb it; on the other hand, if $M_{\text{signal}} \ll M_s$, the shock moves fast enough to escape from the influences of the left region (Fig. 5).

We have neglected the convection speed u_L of the left region only for brevity, but the actual maximum information propagation speed is $(a_L + |u_L|) > a_L \approx 0.44M a_R$. Nevertheless, M_{signal} serves as a good approximation for the lower bound of $(a_L + |u_L|)$. We will come back later to this important number.

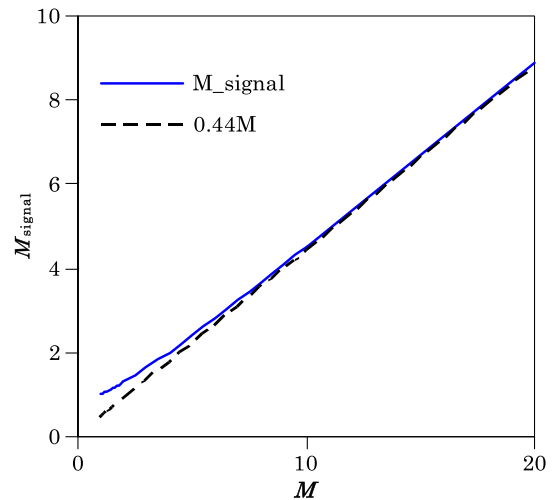


Fig. 6 M_{signal} vs M .

Table 1 Parameters in the present survey

Shock-propagation Mach number M_s	0, 0.5, 1, 1.5, 3, 6, 10
Shock-intensity Mach number M	1.5, 3, 6, 10
Grid perturbation ε_g	0, 1.e-6, 1.e-3
Flux functions	Roe (E-fix), van Leer, AUSM ⁺ -up, SLAU, LDFSS, AUSMPW+, RoeM2, SLAU2

C. Stationary Shock Setup

If a normal shock is placed at a fixed location, Eqs. (2–4) become simpler since $M_s = u_s = 0$. Kitamura et al. [3] varied M from 1.5 to 20 for both 1-D and 2-D stationary shocks, and discovered that the higher the M was, the more likely the shock anomalies emerged, and that such a sensitivity on M died out for $M \geq 6$ ($M_s = 0$). At $M = 6$, for instance, the pre- and postshock states following the Rankine–Hugoniot relation [Eq. (2)] are given as

$$(\rho_L, u_L, v_L, p_L) = (5.2683, 1.1389, 0, 29.881): \text{postshock} \quad (6a)$$

$$= (5.2683, 0.4042a_L, 0, 41.83p_R)$$

$$(\rho_R, u_R, v_R, p_R) = (1, 6, 0, 1/\gamma): \text{pres shock} \quad (6b)$$

The other parameters involved in [3] were shock initial locations ε , flux functions, and spatial accuracy. The shock initial locations included several possible positions “inside” cells. In this study, however, the shock initial location will be fixed precisely on the cell interface ($\varepsilon = 0.0$). Thus, the stationary shock ($M_s = 0$) is merely a special case of the moving shocks, in which the shock takes the only one prescribed location relative to the grid line.

D. Moving Shock Setup by Quirk [1]

Quirk [1] proposed a moving shock test with $M = M_s = 6$ ($u_R = 0$):

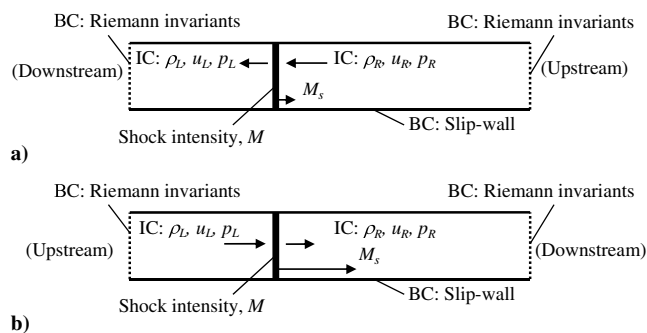


Fig. 7 Typical initial and boundary conditions for a) flows of right-to-left- and b) flows of left-to-right-moving shocks.

$$(r_L, u_L, v_L, p_L) = (5.2683, 4.8611, 0, 29.881): \text{postshock} \quad (7a)$$

$$(\rho_R, u_R, v_R, p_R) = (1, 0, 0, 1/\gamma): \text{pres shock} \quad (7b)$$

In his test, Quirk [1] also introduced another parameter, that is, grid perturbation ε_g as 1.e-3 times the cell size. The perturbation was imposed to the centerline of a 2-D duct through which a shock was propagating as

$$y_{\text{centerline}} = \begin{cases} y_{\text{original}} + \varepsilon_g & (\text{for } i:\text{odd}) \\ y_{\text{original}} - \varepsilon_g & (\text{for } i:\text{even}) \end{cases} \quad (8)$$

The shock can take all the possible locations inside the shock (in contrast to [3]), but may pass through those locations too quickly to be affected by every state. Roberts [9], on the other hand, chose a more slowly moving shock without grid perturbation ($\varepsilon_g = 0$). Thus, as reviewed here, moving shock anomalies have been studied by different researchers from different approaches.

E. Present Setup for General Stationary and Moving Shocks

The parameters involved in the present survey are summarized in Table 1. One can note that included are the choices in Quirk [1] of $(M, M_s) = (6, 6)$ and those in Kitamura et al. [3] of $(M, M_s) = (6, 0)$.

Depending on the magnitudes of M and M_s , flow directions can change, as shown in Fig. 7. The boundary conditions are such that Riemann invariants are conserved along characteristic curves, and either inflow or outflow condition is satisfied according to the flow direction. The grid perturbation parameter ε_g is set to be 1.e-3 times the cell size as default, and 0 and 1.e-6 as alternatives. Flux functions are chosen from three categories [16,21] in terms of robustness against shock and other properties: 1) Roe scheme [18] is known to be vulnerable to carbuncle, even with Harten’s entropy fix (E-fix) [22] (Roe E-fix); 2) van Leer’s FVS [17], AUSMPW+ [30],

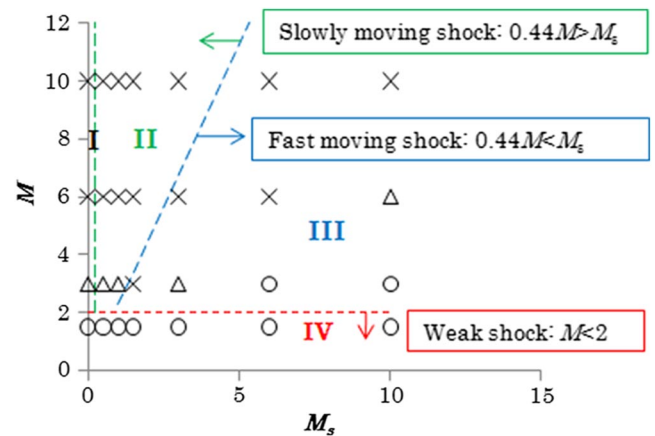


Fig. 9 Moving shock test results ($\varepsilon_g = 1.e-3$), Roe (E-fix).

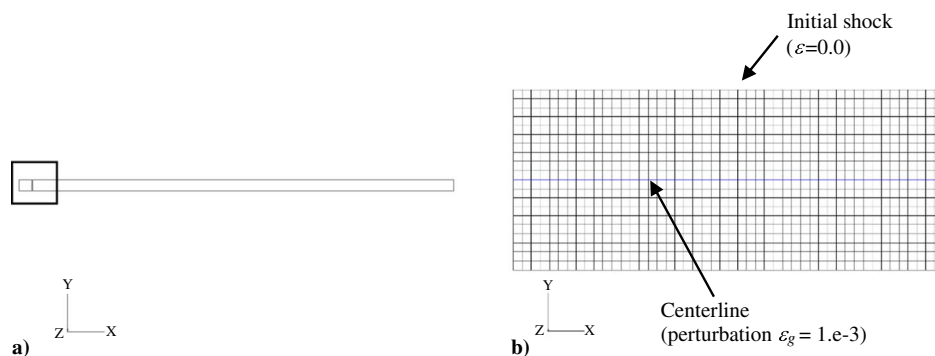


Fig. 8 Moving shock test grid ($\varepsilon_g = 1.e-3$): a) overview and b) close-up view.

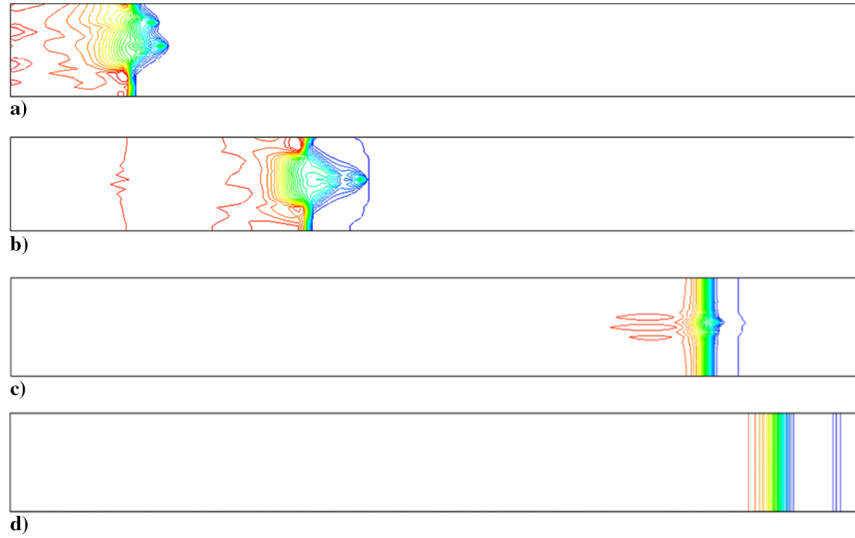


Fig. 10 Moving shock test, $M = 6$ [Roe (E-fix)]: a) $M_s = 0$, b) $M_s = 1.5$, c) $M_s = 6$, and d) $M_s = 10$.

RoeM2 [31], and SLAU2 [27] are almost (but not perfectly) free from such anomalies; and 3) AUSM⁺-up [19], LDFSS [29], and SLAU [20] are in the middle. The computations are run for 400 steps with CFL = 0.4 (Fig. 8). Following Quirk [1], the grid consists of 800×20 cells. The initial shock is placed at the interface between the 25th and 26th cells ($\varepsilon = 0.0$).

IV. Numerical Results

A. Effects of Two Mach Numbers M_s and M

Figure 9 summarizes Roe (E-fix) solutions for $M_s = 0, 0.5, 1.0, 1.5, 3.0, 6.0$, and 10.0 ($\varepsilon_g = 1.e - 3$), according to the following criteria:

- 1) $\circ$: stable. L1-norm of v -velocity residual remained the same order from the beginning.
- 2) Δ : oscillatory, but no carbuncle. L1-norm of v -velocity residual grew over one order.
- 3) $\times$: carbuncle (total breakdown) or divergence. The shock oscillated spatially over four cells or more in the freestream direction if carbuncle.

Here, we used the L1-norm of velocity component v (crossflow velocity component), which is supposed to be zero in a theoretical sense, as a measure of (multidimensional) shock anomalies. As seen, the stable and carbuncle solutions exist, but an increase of M_s allows higher M . Very-fast-moving shocks, such as $(M, M_s) = (3, 10)$, showed a stable solution, whereas a mildly fast-moving shock $(M, M_s) = (3, 2)$ was unstable. Typical visualized results are shown in Fig. 10 for $M = 6$; $M_s = 0, 1.5, 6$ (correspond to Quirk [1]), and 10. This figure also demonstrates that the faster the shock propagates (the higher the M_s), the more stable it is.

Figure 11 shows L1-norms of v residuals, whose growth in time indicates multidimensional instability. As seen from Fig. 11a, the

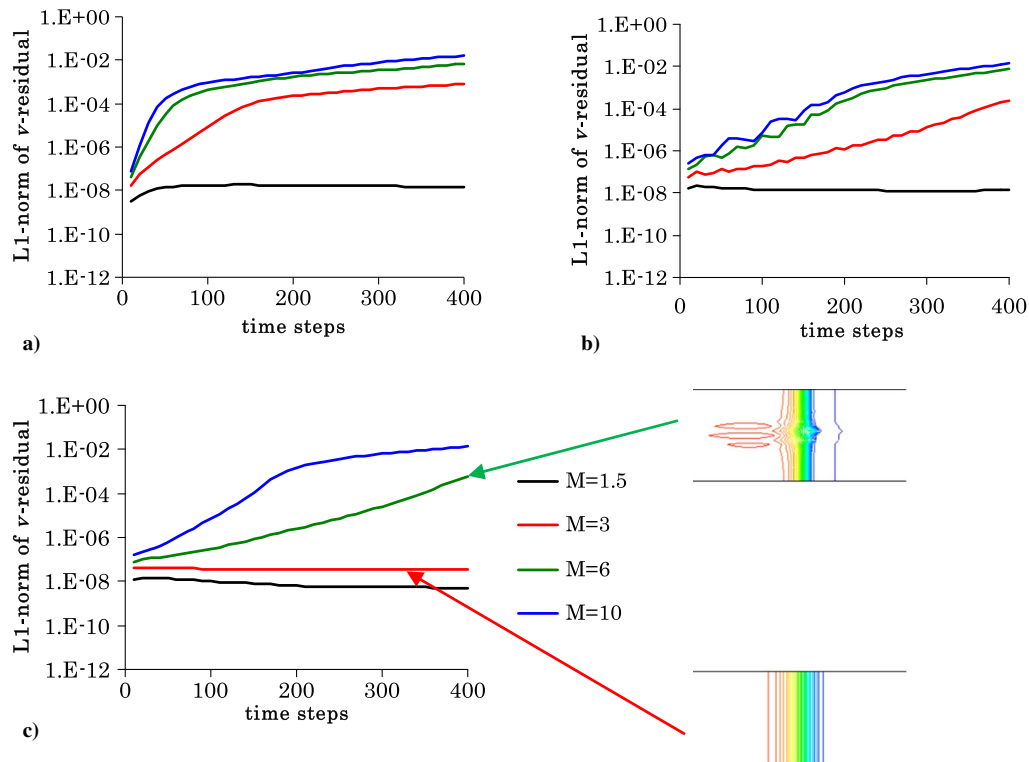


Fig. 11 L1-norm histories of v -velocity residual: $M = 1.5, 3, 6$, and 10 [Roe (E-fix)]: a) $M_s = 0.0$, b) $M_s = 0.5$, and c) $M_s = 6$.

$M_s = 0$ (stationary) shocks exhibit such unstable growth except for the $M = 1.5$ (the weakest) case ($M < 2$: the weak shock); the second weakest $M = 3$ shock shows slight delay of the growth of instability when $M_s = 0.5 < 0.44M$ (slowly moving) in Fig. 11b; the two weakest shocks, $M = 1.5$ and 3, are stable for $M_s = 6 > 0.44M$ (fast moving) in Fig. 11c.

B. Effects of Flux Functions

Figure 12 shows the summarized results for AUSM⁺-up, SLAU, van Leer, LDFSS, AUSMPW+, RoeM2, and SLAU2. Among them, AUSMPW+ and van Leer's FVS [17] are stable except for the two

cases of van Leer, in which v appears, but the shock itself looks symmetry (indicated by solid circles). (Nevertheless, this does not mean that they are always stable under any conditions. On meshes with high aspect ratio, AUSMPW+ may be unstable [16].) SLAU2 and RoeM2 are almost always stable, with only a few cases of slight v -velocity growth. This observation appears to contradict with [27], in which SLAU2 showed the best stability against shock anomalies, whereas AUSMPW+ did not [3]. In [3] and [27], however, no grid perturbations were introduced. Thus, for perturbed grids, a multidimensional pressure weight function in AUSMPW+ may have acted favorably, whereas SLAU2, equipped with no such a term, has been affected by the grid under particular setups.

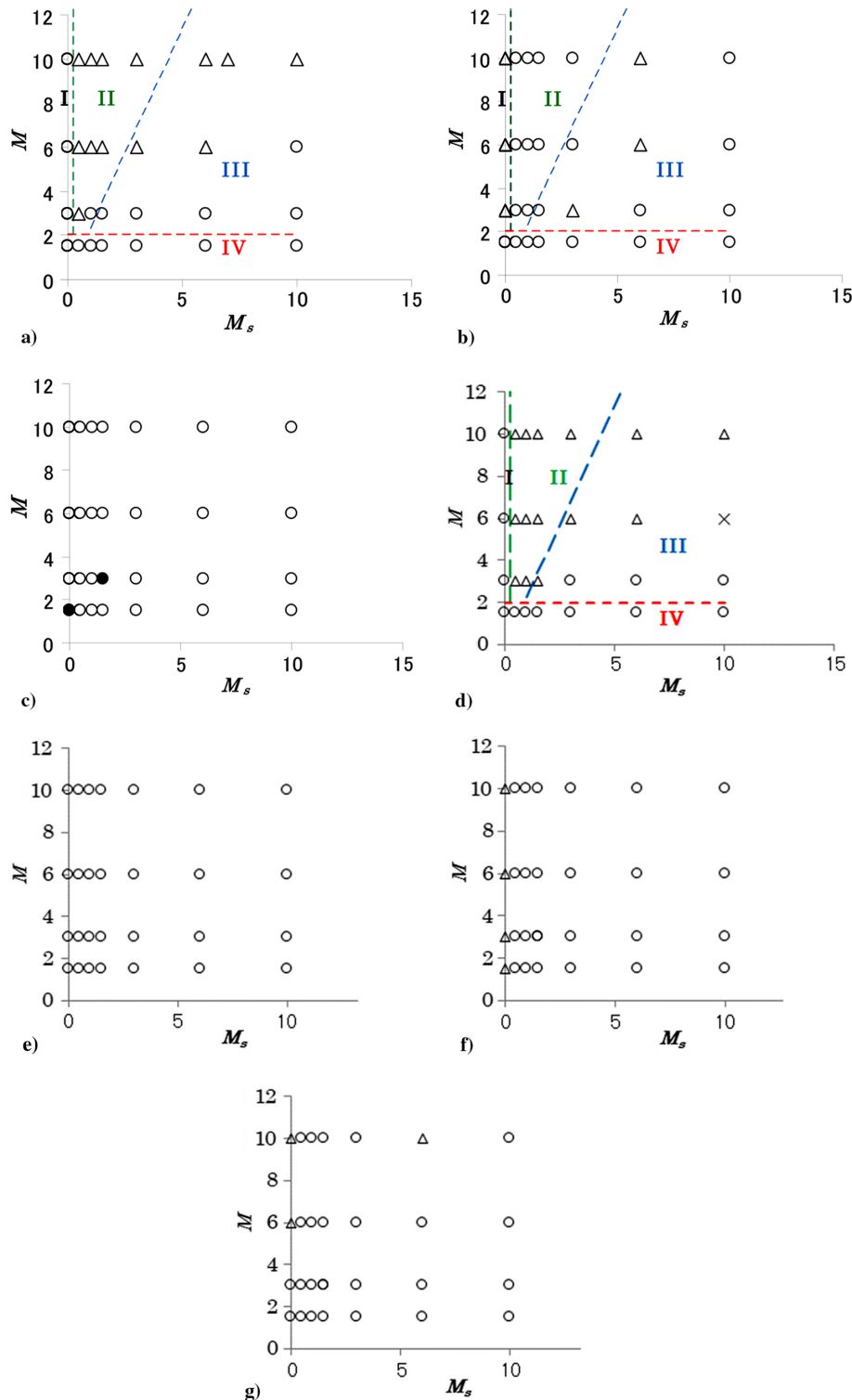


Fig. 12 Moving shock test results ($\epsilon_g = 1.e - 3$): a) AUSM⁺-up, b) SLAU, c) van Leer, d) LDFSS, e) AUSMPW+, f) RoeM2, and g) SLAU2.

Furthermore, if the stable criterion is relaxed to allow one order growth of v velocity (i.e., Two orders or more are called *oscillations*), the solutions of SLAU2 will be all relabeled as stable.

AUSM⁺-up, SLAU, and LDFSS showed mixed solutions of stable and oscillatory. Nevertheless, all these are more robust than Roe (E-fix), and again, the faster the shock (the higher the M_s), the more stable the shock for AUSM⁺-up (except for a stationary shock), for instance.

Figures 13–15 confirm that, at $M = 6$, all the solutions look stable for AUSM⁺-up, SLAU, and van Leer [17]. (No oscillations are visible from these figures.) However, showing these figures only may misguide the readers. Having seen Fig. 12 together, we are aware of the fact that a different (M, M_s) set yielded a different solution, and that we need such a broad picture to examine the performance of the flux functions. In other words, only one particular shock speed (e.g., Quirk's test [1]) may not be enough to support the robustness of the flux function against shocks.

In fact, the stationary shocks ($M_s = 0$) behaved totally differently from the moving ones (Figs. 11a and 11b). This is understandable considering the fact that each flux function has its own “comfortable” shock location relative to grid lines [3]. In AUSM⁺-up cases, for instance, the stationary shocks located exactly at the cell interface showed no sign of shock anomalies (Figs. 12a and 13a).

However, on the contrary to the AUSM⁺-up, the stationary shocks are less stable than the moving counterparts in SLAU (Fig. 12b): in very general, SLAU showed more-stable moving shocks compared to AUSM⁺-up, but this is the opposite for stationary shocks at cell interface. Furthermore, because there are many stable moving shocks in Fig. 12b, it is hard to find any tendency according to (M, M_s) combinations, for this particular flux function.

The final flux function focused on is van Leer's FVS [17]. This flux is known to be very robust against the (stationary) shock anomalies, and this still holds for moving shocks according to Figs. 12c and 15. Thus, the stable combinations of (M, M_s) and the overall robustness against the shock differ from one flux function to another. Nevertheless, the $0.44M$ (approximate signal speed) is useful to categorize the results for simple estimations as follows:

1) Group 1 (stationary shock: $M_s = 0, M \geq 2$): because the shock did not cross any cells, there should be no influence from the grid perturbation. Each flux function has its own comfortable shock location relative to grid lines [3], and AUSM⁺-up, LDFSS, and AUSMPW+ showed no instability regardless of shock intensity M .

2) Group 2 (slowly moving shock: $0 < M_s < 0.44M, M \geq 2$): because the shock crosses a perturbed grid with a long time (during which the instabilities can grow) and because numerical noise from the left (downstream) region (Fig. 7a) can affect the shock itself (as explained in Sec. III.B), the solution is most likely to exhibit



Fig. 13 Moving shock test, $M = 6$ (AUSM⁺-up): a) $M_s = 0$, b) $M_s = 1.5$, c) $M_s = 6$, and d) $M_s = 10$.



Fig. 14 Moving shock test, $M = 6$ (SLAU): a) $M_s = 0$, b) $M_s = 1.5$, c) $M_s = 6$, and d) $M_s = 10$.

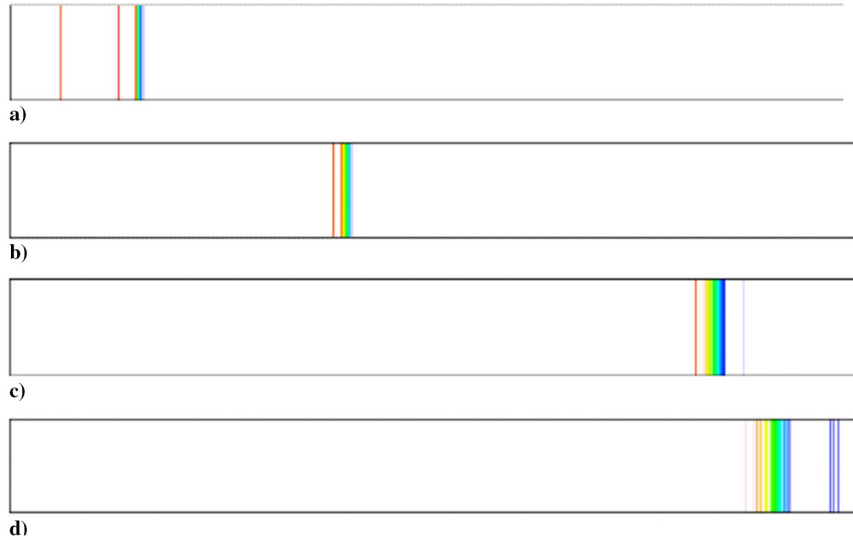


Fig. 15 Moving shock test, $M = 6$ (van Leer): a) $M_s = 0$, b) $M_s = 1.5$, c) $M_s = 6$, and d) $M_s = 10$.

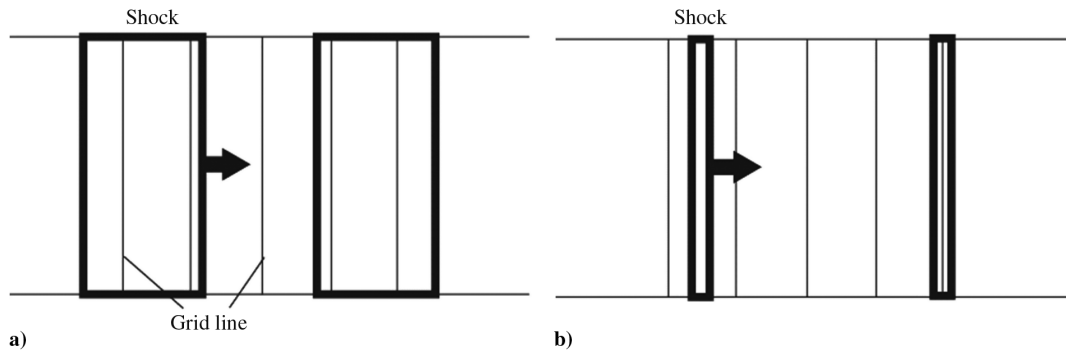


Fig. 16 Schematics of moving shocks captured by a) SLAU and b) Roe.

carbuncle. In Roe (E-fix), for instance, the solutions are usually carbuncle with a few exceptions of slight oscillation cases.

3) Group 3 (fast-moving shock: $0.44M < M_s$, $M \geq 2$): because the shock crosses a perturbed grid with a short time and numerical noise from the left (downstream) region hardly reaches the shock itself, the solution is less likely to exhibit anomalies than group 2. The solutions are mixed with stable, oscillatory, and carbuncle cases. Although the border between groups 2 and 3 is not clear (because it is based on $0.44M$, approximated signal speed), group 3 tends to be more stable than group 2 (except for SLAU).

4) Group 4 (weak shock: $M < 2$): the shock intensity is not strong enough to provoke carbuncle. The solutions are usually free from such anomalies.

The reason why SLAU behaved differently would be related to its inherent shock-capturing feature [21]. Roe and AUSM⁺-up capture a shock sharply (i.e., zero or one cell), whereas SLAU (as well as SLAU2 and van Leer) represents it with a few cells (i.e., one or more). Thus, SLAU is less sensitive to the moving shock speed, because it already contains the (numerical) shock internal structure (Fig. 16a), as opposed to Roe or AUSM⁺-up (Fig. 16b). AUSMPW⁺ and RoeM2, having a multidissipation term to successfully stabilize the shock, are also insensitive to the shock-propagation speed.

C. Effects of Grid Perturbation ϵ_g

Finally, Fig. 17 shows the effects of the grid perturbation ϵ_g , which is reduced from the default value of $1.e-3$ to $1.e-6$, and even 0 (Solid black circles stand for rapid growth of v , although still very small value [$\mathcal{O}(10^{-12})$ or smaller]). As seen, the solutions (common to both $\epsilon_g = 1.e-6$ and 0) look more stable: only the unstable solutions are found at $(M, M_s) = (10, 0)$ and $(6, 0)$ (i.e., stationary

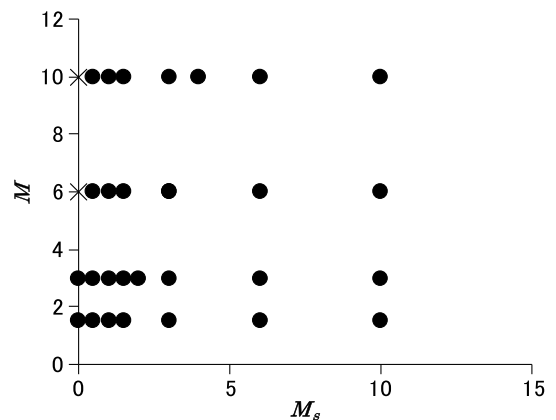


Fig. 17 Moving shock test results [Roe (E-fix), $\epsilon_g = 1.e-6$ or 0].

shocks); otherwise, the velocity v is very small (within $1.e-12$). Its growth implies potential carbuncle at a very later stage (i.e., after a very long computation).

Nevertheless, within the present framework, neither of $\epsilon_g = 1.e-6$ nor 0 showed a clear sign of carbuncle, and no distinct difference was observed between them. Thus, the grid perturbation ϵ_g has a minor effect on the results as long as it is taken to be small enough. On the other hand, in Quirk [1], $\epsilon_g = 1.e-3$ was chosen, which was large enough to trigger the carbuncle and other shock anomalies at an early stage of computations. Thus, $\epsilon_g = 1.e-3$ is found to be a good value to check the occurrence of shock anomalies in a realistic time.

V. Conclusions

As a systematic and thorough benchmark, numerical experiments for a variety of shock-intensity Mach number M and shock-propagation Mach number M_s , with different flux functions and different levels of grid perturbations, have been carried out. According to the difference between the flux functions

1) SLAU, van Leer, AUSMPW+, RoeM2, and SLAU2: they are not affected by shock-propagation speeds, because the shock is captured over a few cells, and/or a multidimensional dissipation term successfully stabilizes the shock.

2) Roe, AUSM⁺-up, and LDFSS: they are easily affected by shock-propagation speeds, because they tend to capture the shock sharply (within at most one cell).

Then, for the latter fluxes, the numerically captured shock waves are categorized into the four groups having different degrees of robustness against shock anomalies according to M and M_s :

1) Stationary shocks ($M_s = 0$; $M \geq 2$): they are either more or less stable than moving ones, depending on the flux functions.

2) Slowly moving shocks ($0 < M_s \leq 0.44M$; $M \geq 2$): this is the most severe case, because they are affected by both grid perturbation and numerical noise from the left (downstream) region.

3) Fast-moving shocks ($0.44M < M_s$; $M \geq 2$): they are less likely to exhibit shock anomalies, because they are less influenced by the left (downstream) region.

4) Weak ($M < 2$) shocks: they are usually stable.

Thus, only a particular set of (M , M_s) [which is only a fraction of the broader (M , M_s) charts presented herein] cannot support the stability of a flux function enough. Finally, the present default choice of grid perturbation following Quirk [1] ($1.e - 3$ the cell size) turned out to be big enough to trigger the shock anomalies at earlier stages.

Now, the ultimate goal has been set: the authors will explore the universal flux function, which is always robust against shock anomalies for all the (M , M_s), while applicable to a contact discontinuity capturing, hypersonic wall heating, and all-speed flow computations. Such a study is left as a future work, but SLAU2 used with the minmod slope limiter is recommended, for the time being [21].

Acknowledgments

This work was partially conducted while Keiichi Kitamura was under supervision of Eiji Shima at the Japan Aerospace Exploration Agency, supported by the Japan Society for the Promotion of Science. Tsutomu Saito, Muroran Institute of Technology, Japan, gave the authors a helpful comment. The authors thank their cooperation.

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